

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



232/1

Physics (Theory)

Paper 1

Time: 2 Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES:

- (a) Write your Name and Index Number in the spaces provided.
- (b) Sign and write the date of examination in the spaces provided above.
- (c) This paper consists two sections A and B.
- (d) Answer all questions in Section A and B in the spaces provided.
- (e) All working **MUST** be clearly shown.
- (f) Mathematical tables and electronic calculators may be used.

FOR EXAMINER'S USE ONLY

Section	Question	Max. Score	Candidate's Score
A	1 – 12	25	
B	13	10	
	14	10	
	15	11	
	16	13	
	17	11	
Total Score		80	

SECTION A (25 MARKS)

Answer ALL questions in this section in the spaces provided

1. The **figure 1**, shows a micrometer screw gauge that has a zero error of $+0.02$. State the actual reading of the micrometer screw gauge. (1mk)

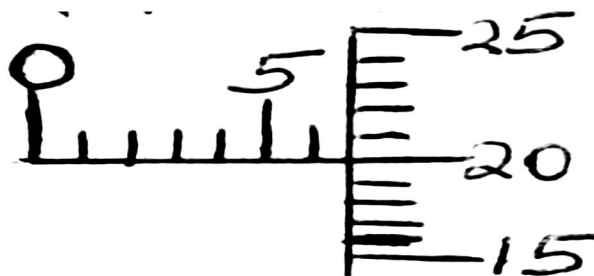


Fig. 1

2. In the **figure 2**, the U-tube contains two immiscible liquids P and Q. If the density of Q is 900kg/m^3 and that of P is 1200kg/m^3 , calculate the height of liquid Q. (3mks)

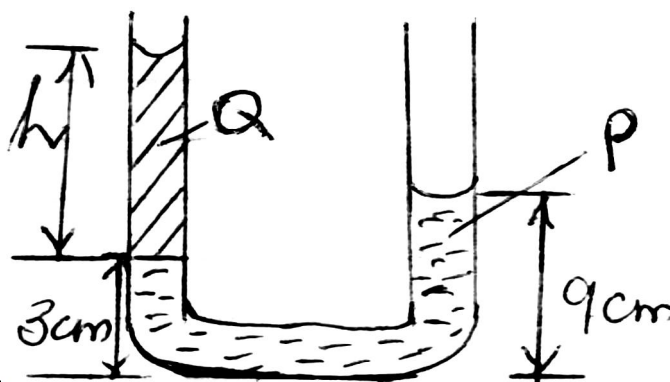


Fig. 2

3. A force of 10N towards the right hand side and 6N towards the left hand side acts upon a body. What is the resultant force? (2mks)

4. A trolley of mass 1.5kg moving with a velocity of 1.2ms^{-1} collides inelastically with a second trolley of mass 0.5kg moving in the opposite direction with a velocity of 0.2ms^{-1} .

(a) What is an inelastic collision? (1mk)

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(b) Determine the velocity of the trolleys after collision. (2mks)

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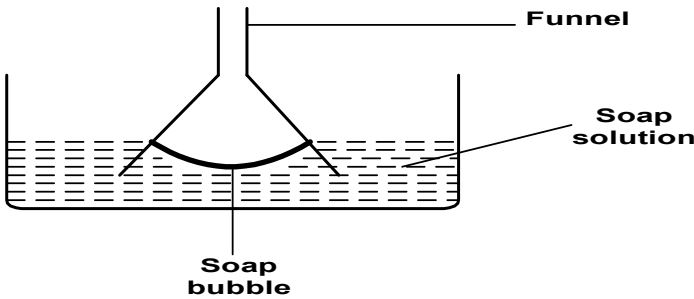
5. a) What is surface tension? (1mark)

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b) Figure 3, shows a funnel dipped into a liquid soap solution.

Fig. 3



Explain what happens to the soap bubble when the funnel is removed. (1mark)

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6. **Figure 4**,shows a clinical thermometer which is not graduated.

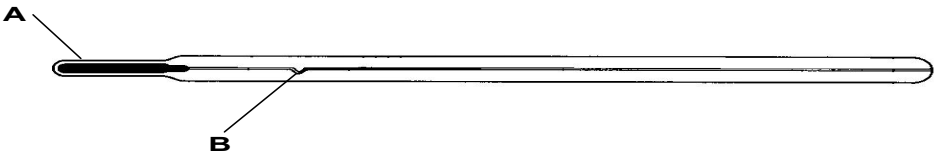


Fig. 4

a) Name the parts indicate with letters: A and B (1 mark)

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b) Mark the appropriate scale range in degrees Celsius (1 mark)

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7. **Figure 5**, shows air flowing through a pipe of non-uniform cross-sectional area. Two pipes A and B are dipped into liquids as shown.

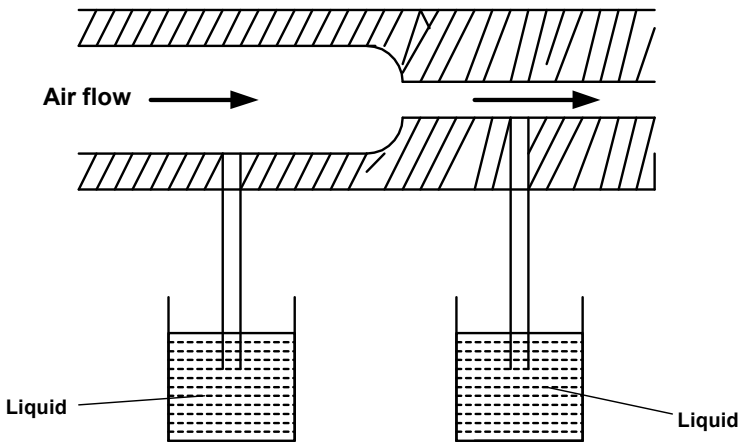


Fig. 5

B

A

a) Indicate the levels of the liquids in Pipe A and pipe B. (1 mark)

b) Explain your answer in 7 (a) above. (1 mark)

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8. The **figure 6**, shows a flask fitted with a glass tube dipped into a beaker containing water at room temperature. The cork fixing the glass tube is tight.

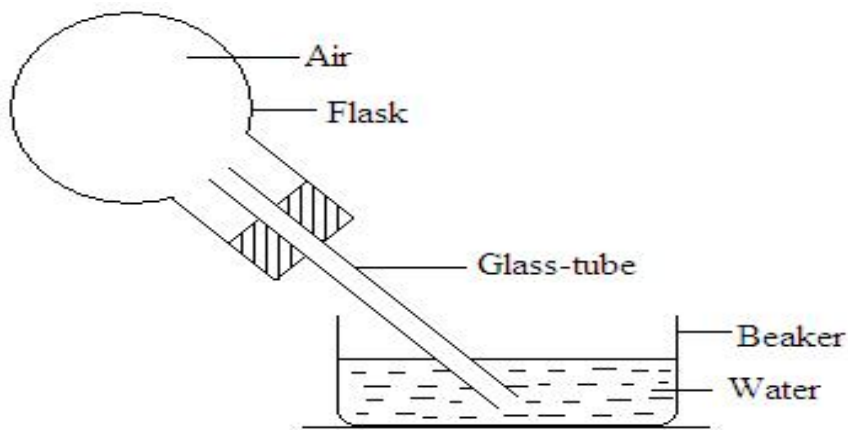


Fig. 6

State with reason what would be observed if cold water is poured on to the flask. **(2marks)**

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9. The **figure 7**, shows three identical springs which obey Hooke’s law.

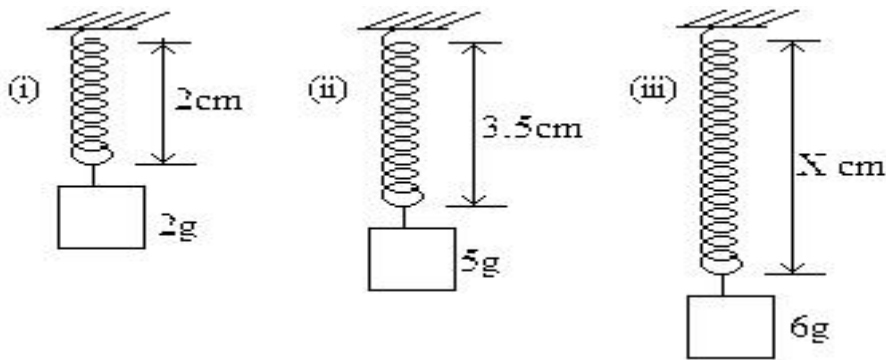


Fig.7

Determine the length Xcm. **(3 marks)**

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10. Using the idea of particles, explain why the pressure inside the tyre is increased when it is pumped up
(2marks)

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11. Give one fact which shows that heat from the sun does not reach the earth surface by convection.
(2marks)

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12. (a) Give a reason why water is not suitable as a barometric liquid. (1 mark)

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(b) Explain why a lift pump is unable to raise water from a borehole where the level of water is 20m below the ground level. (1 mark)

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SECTION B (55 Marks)

Answer ALL questions in this section in the spaces provided.

13. a) A force of 7.5N stretches a certain spring by 5 cm. Calculate the work done in stretching this spring by 8.0 cm. (3 marks)

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b) **Figure 8**, shows a cross-section of a handle of a screw jack 70 cm long. The pitch of the screw is 0.8 cm

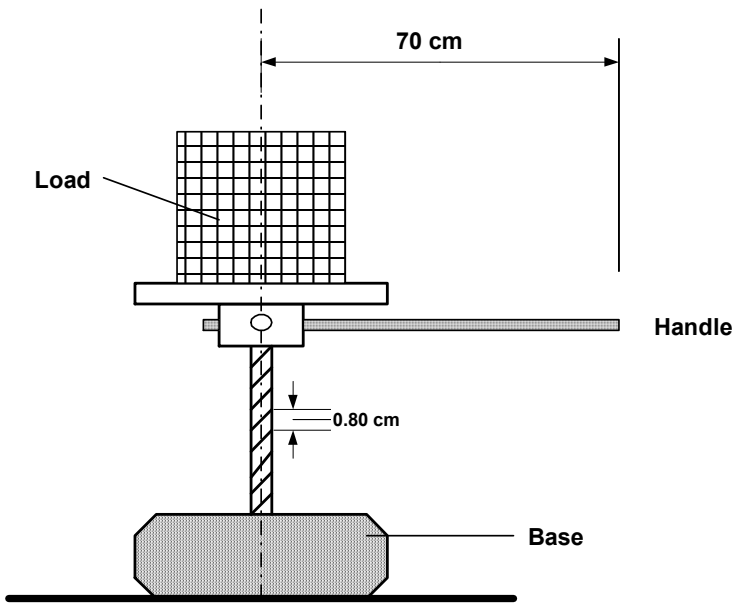


Fig. 8

Given that the efficiency is 65%, calculate:

i) the velocity ratio of the system (2 marks)

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iii) the mechanical advantage of the screw jack. (2 marks)

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iii.) Sketch a graph of efficiency against Load (1 mark)

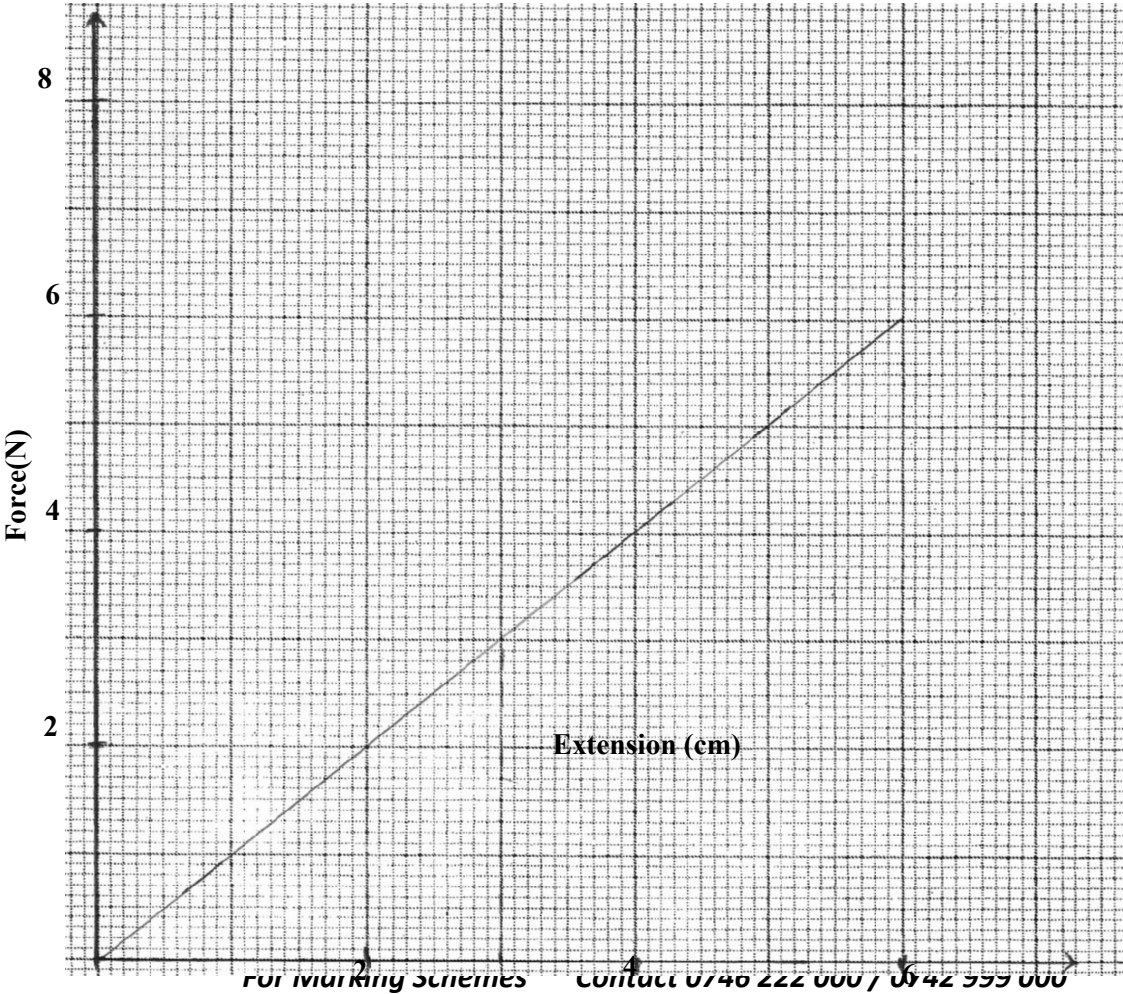
iv.) Draw a single moving pulley with a velocity ratio of 2. (2 marks)

14. (a) State Hooke’s law (1mark)

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(b) The graph provided is of force (y-axis) against extension.



(g) From the graph determine the work done in stretching spring by 3cm **(3marks)**

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(ii) Use the graph to determine the spring constant. Give your answer in SI units **(3marks)**

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(iii) State three factors that affect the spring constant **(3marks)**

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15. Define latent heat of vaporization. **(1 mark)**

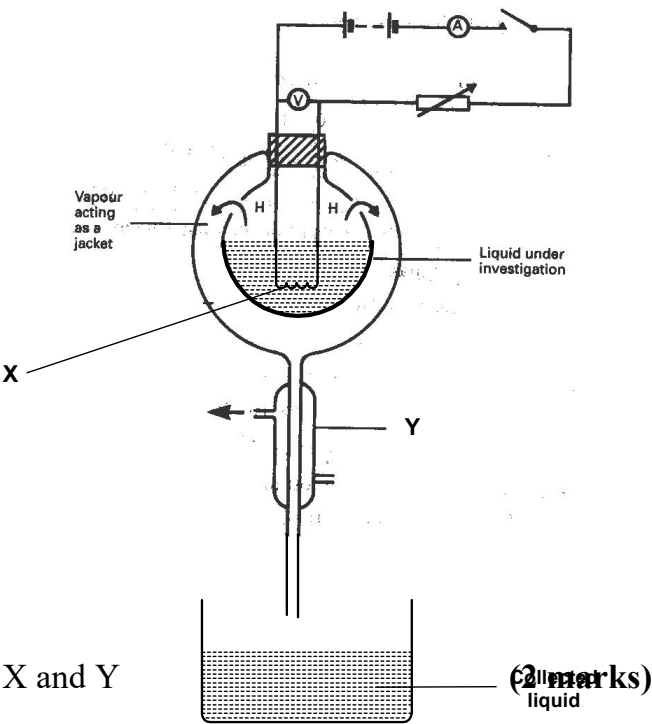
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b) **Figure 9**, shows a set up by a student to determine the specific latent heat of vaporization of a liquid.

Fig. 9



i) **Identify** the parts labelled X and Y (2 marks)

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ii) **State** the measurements that should be taken. (2 marks)

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iii) **Describe** how the set up can be used to determine the specific latent heat of vaporisation of the liquid. (5 marks)

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iv) What is the purpose of the condenser? (1 mark)

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16. a)i) State Archimedes’s Principle. (1 mark)

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(ii) An object weighs 1.04N in air, 0.64N when fully immersed in water and 0.72N when fully immersed in a liquid. If the density of water is 1000 kg m^{-3} , find the density of the liquid.

(4 marks)

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b) i) State the law of floatation (1 mark)

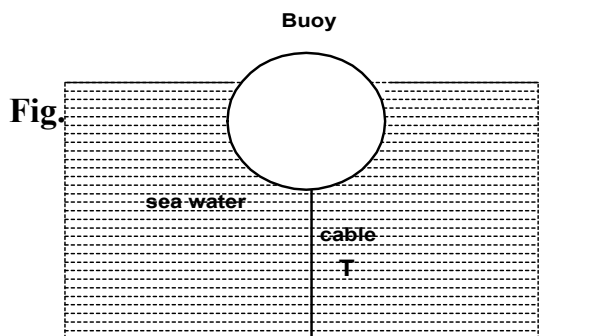
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ii) Give a reason why a steel rod sinks in water while a ship made of steel floats on water.(1 mrk)

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iii) Draw a clearly labelled diagram of a common hydrometer, which is suitable for measuring the densities of liquids varying between 1.0 and 1.2 g cm^{-3} . **(2 marks)**

iv) **Figure 10**, shows a buoy, B, of volume 40 litres and mass 10 kg . It is held in position in sea water of density 1.04 g cm^{-3} by a light cable fixed to the bottom so that $\frac{3}{4}$ of the volume of the buoy is below the surface of the sea water. Determine the tension T in the cable. **(4 marks)**



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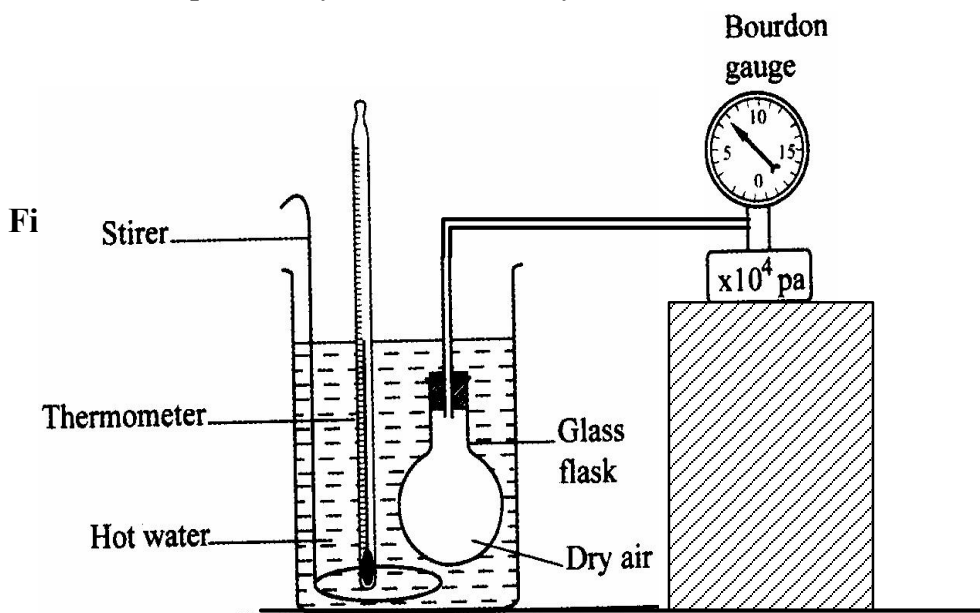
17. a) State Pressure Law

(1 mark)

b) State one basic assumption of the kinetic theory of gases.

(1marks)

c) Figure 11, shows a set up that may be used to verify Pressure law.



i. State the measurements that may be taken in the experiment.

(2 marks)

ii) Explain how the measurement in (i) above may be used to verify Pressure law.

(4 marks)

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d)) A bicycle tyre is pumped to a pressure of 2.2×10^5 pa at 23°C . After a race the pressure is found to be 2.6×10^5 pa. Assuming the volume of the tyre did not change, what is the temperature of the air in the tyre. **(3 marks)**

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